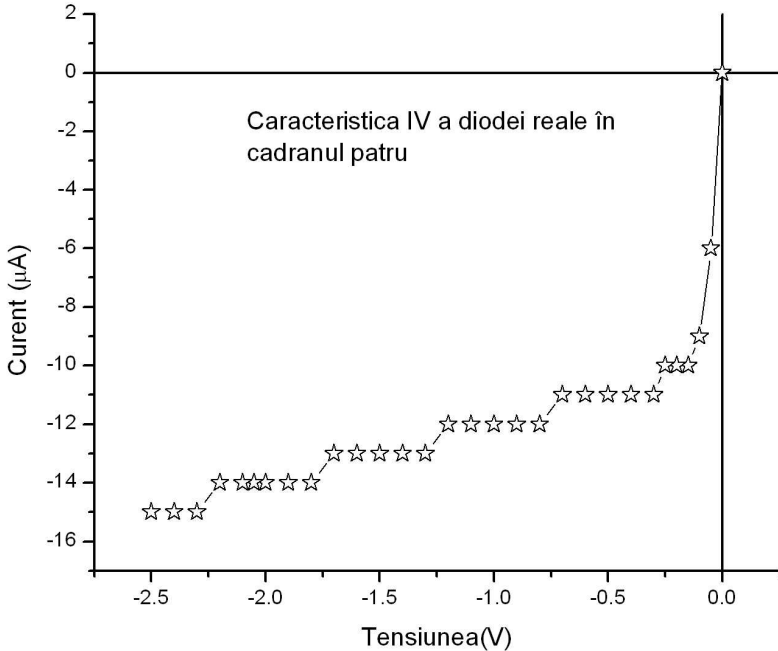
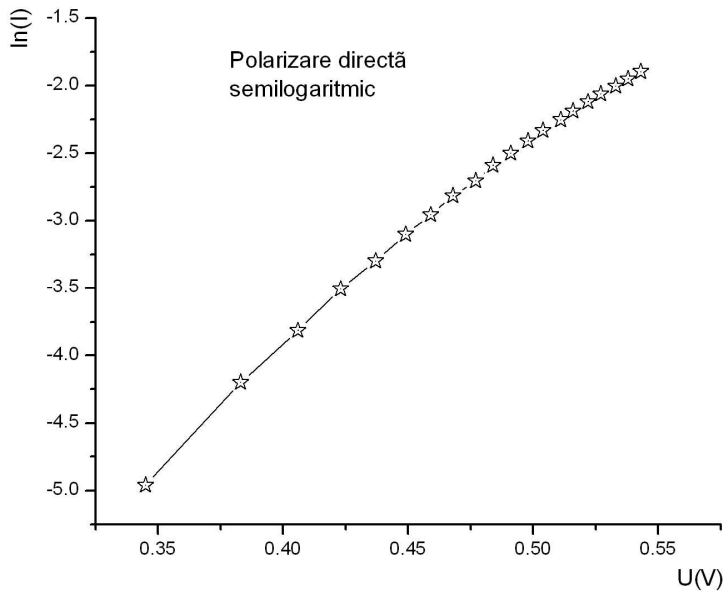


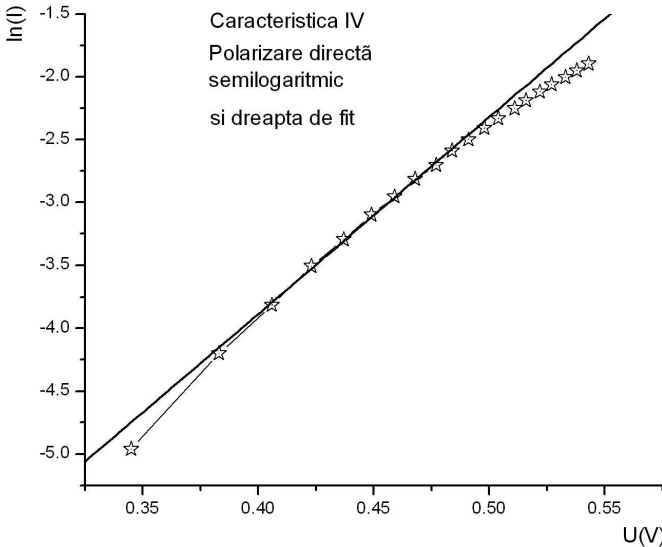
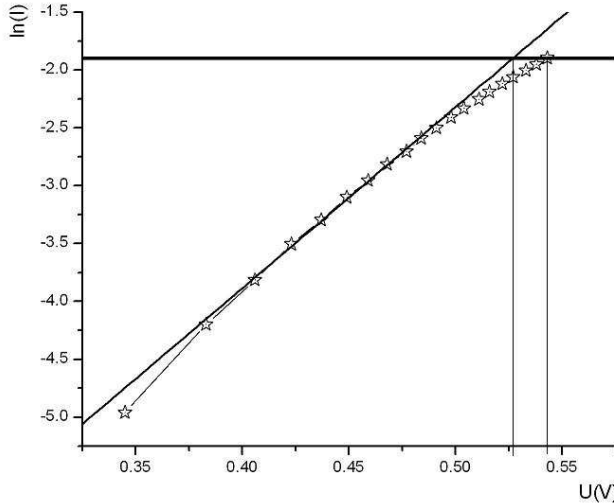
Evaluation Sheet

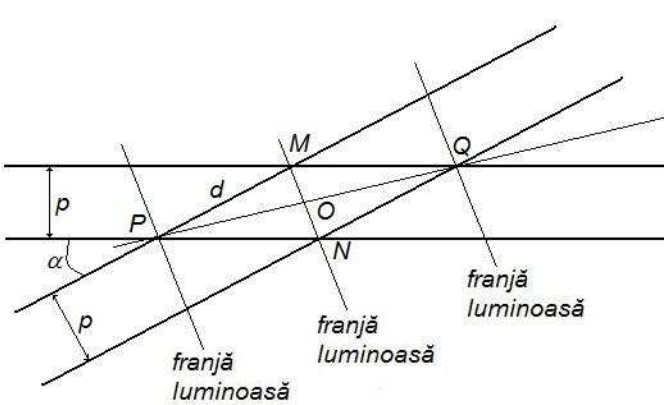
Any other correct solution will be evaluated accordingly

No.	<i>Experimental problem 1</i>	Points
1.	For: $I = -I_0 \cdot + \frac{V}{R}$ 0.4p	0.4p
2.	For:  0.6p	0.6p
3.	For: Linear expression of the $I - V$ characteristic for reversed polarity $I = \frac{V}{R} - I_0$ 0.2p $a = 1/R$ 0.2p $b = -I_0$ 0.2p	0.6p
4.	For: correctly filling in the Table 3 0.8p	0.8p

5.	For: the slope of the fitting line $a = \frac{10 \sum_{i=1}^{10} U_i \cdot I_i - \sum_{i=1}^{10} U_i \cdot \sum_{i=1}^{10} I_i}{10 \cdot \sum_{i=1}^{10} U_i^2 - \left(\sum_{i=1}^{10} U_i \right)^2}$ $a = \frac{10 \cdot 298.8 - 20.95 \cdot 142}{10 \cdot 44.49 - (20.95)^2} = \frac{13.1}{5.99} \frac{\mu A}{V} = 2.18 \times 10^{-6} \frac{A}{V}$ the slope-intercept of the fitting line $b = \frac{\sum_{i=1}^{10} U_i^2 \cdot \sum_{i=1}^{10} I_i - \sum_{i=1}^{10} U_i \cdot \sum_{i=1}^{10} U_i \cdot I_i}{10 \cdot \sum_{i=1}^{10} U_i^2 - \left(\sum_{i=1}^{10} U_i \right)^2}$ $b = \frac{44.49 \cdot (-142) - (20.95) \cdot 298.8}{5.99} = \frac{-57.72}{5.99} \mu A = -9.6 \mu A$	0.8p 0.4p 0.4p
6.	For: the correlation coefficient for the presumed linear dependency $G = \frac{10 \cdot \sum_{i=1}^{10} U_i \cdot I_i - \sum_{i=1}^{10} U_i \cdot \sum_{i=1}^{10} I_i}{\sqrt{\left 10 \cdot \sum_{i=1}^{10} U_i^2 - \left(\sum_{i=1}^{10} U_i \right)^2 \right \cdot \left 10 \cdot \sum_{i=1}^{10} I_i^2 - \left(\sum_{i=1}^{10} I_i \right)^2 \right }}$ $G = \frac{10 \times 298.8 - 20.95 \times 142}{\sqrt{5.99 \times 36}} = \frac{13.1}{14.6} \cong 0.90$	0.4p 0.4p
7.	For: $I_0 = 9.6 \mu A$ $R = \frac{1}{a} \cong 460 k\Omega$	0.8p 0.4p 0.4p
8.	For: $I = I_0 \cdot e^{\frac{q(V - rI)}{\eta \cdot k_B \cdot T}}$	0.4p 0.4p
9.	For: $\ln(I) = \ln(I_0) + \frac{q(V - rI)}{\eta k_B T}$ for experimental data within the range $V > rI$ $\ln(I) = \ln(I_0) + \frac{qV}{\eta k_B T}$	1.0p 0.2p 0.2p

	 0.6p	
10.	For: correctly filling in table 4 0.8p	0.8p
11.	For: the slope of the fitting line $a = \frac{10 \sum_{i=1}^{10} U_i \cdot \ln(I_i) - \sum_{i=1}^{10} U_i \cdot \sum_{i=1}^{10} \ln(I_i)}{10 \cdot \sum_{i=1}^{10} U_i^2 - \left(\sum_{i=1}^{10} U_i \right)^2}$ $a = \frac{10 \cdot (-12.493) + (4.857) \cdot (25.849)}{10 \cdot (2.363) - (4.857)^2} = \frac{0.6185}{0.0395} = 15.66$ the slope-intercept of the fitting line $b = \frac{\sum_{i=1}^{10} U_i^2 \cdot \sum_{i=1}^{10} \ln(I_i) - \sum_{i=1}^{10} U_i \cdot \sum_{i=1}^{10} U_i \cdot \ln(I_i)}{10 \cdot \sum_{i=1}^{10} U_i^2 - \left(\sum_{i=1}^{10} U_i \right)^2}$ $b = \frac{(2.363) \cdot (-25.849) + (4.857) \cdot (12.493)}{0.0395} = \frac{-0.403}{0.0395} = -10.15$ 0.4p	0.8p
12.	For: the correlation coefficient for the presumed linear dependency $G = \frac{10 \cdot \sum_{i=1}^{10} U_i \cdot \ln(I_i) - \sum_{i=1}^{10} U_i \cdot \sum_{i=1}^{10} \ln(I_i)}{\sqrt{\left 10 \cdot \sum_{i=1}^{10} U_i^2 - \left(\sum_{i=1}^{10} U_i \right)^2 \right \cdot \left 10 \cdot \sum_{i=1}^{10} (\ln(I_i))^2 - \left(\sum_{i=1}^{10} \ln(I_i) \right)^2 \right }}$ $G = \frac{0.6185}{\sqrt{0.04 \cdot 8.4}} \cong 1$ 0.4p	0.4p

13.	For:  Caracteristica IV Polarizare directă semilogaritmă și dreaptă de fit	0,6p
14.	For: $\begin{cases} \eta = \frac{q}{a \cdot k_B T} \\ I_0 = e^b \end{cases}$ 0,4p $\begin{cases} \eta = \frac{38,64}{15,66} = 2.4 \\ I_0 = 3.7 \times 10^{-5} \text{ A} \cong 30 \mu\text{A} \end{cases}$ 0,4p	0.8p
15.	 0.6p $\Delta V = rI$ $0.02 = r \cdot 0.15$	0.6p
16.	For: $r \cong 0.2 \Omega$	0.2p
TOTAL		10p

No.	<i>Experimental problem 2</i>	Points
A.	For: the value of the period of the Moiré pattern: 23 u.a. 1.0p the value of the period of the grid marked “Rigla, raportor, grila 2”: $p = \frac{1}{1/0.76 \pm 1/23}$ 0.5p	1.5p
B.	the value of the period of the Moiré pattern: 23 u.a. 1.0p the value of the period of the grid marked “Rigla, raportor, grila 3”: $p = \frac{1}{1/0.76 \pm 1/23}$ 0.5p	1.5p
C.	For: correct commentary regarding the possibility for a Moiré pattern to appear as a consequence of the usage of the grids indicated 1.0p	1.0p
D.	For: the value of the period of the grid marked “Rigla, raportor, grila 2”: 0.79 u. a. 1.0p the value of the period of the grid marked “Rigla, raportor, grila 3”: 0.73 u. a. 1.0p	2.0p
E.	For:  0.5p	2.0p
	$PO = d \cdot \cos\left(\frac{\alpha}{2}\right)$ 0.5p	
	$PO = p \cdot \frac{\cos\left(\frac{\alpha}{2}\right)}{\sin \alpha}$ 0.5p	
	$PO = \frac{p}{\sqrt{2}} \cdot \sqrt{\frac{1 + \cos \alpha}{\sin^2 \alpha}}$ 0.5p	

F.	For: correctly filling in Table 1 1.0p	1.0p
G.	For: correctly filling in Table 2 1.0p	1.0p
<i>TOTAL</i>		10p